JY-CRPG-BENCH: A LONG-HORIZON BENCHMARK FOR VISION-LANGUAGE MODELS IN A CHINESE OPEN-WORLD RPG

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ABSTRACT

A long-horizon task requires a model to perceive its environment, decide for itself what is worth doing, and hold one plan across hours of dependent decisions. We present JY-CRPG-BENCH, a benchmark that places a vision-language model inside *Heroes of Jin Yong*, a 1996 open-world role-playing game run under DOS emulation, and scores it against the ending the game defines, which takes a first-time human player tens of hours. The model sees a 320×200 frame and presses keys. The game speaks only Traditional Chinese, draws its world in an isometric projection in which no key moves straight up the screen, and states its objectives only inside its fiction. Progress is read from the memory of the emulated machine, from the save the game writes and from the published replay, so a ladder of verified milestones orders the field before any model finishes, and completion time orders it after. The benchmark, its replays and every session are public at https://github.com/hanxiao/jy-crpg-bench.

1 INTRODUCTION

Frontier models prove competition mathematics and write working compilers, yet the tasks they complete reliably are still measured in hours of human work (Kwa et al., 2025). A long horizon is a property of the task, the number of dependent steps its completion requires (Chen et al., 2026), and the benchmarks that measure it have so far run in text, as business simulations (Backlund & Petersson, 2025; Fan et al., 2026) or interactive fiction (Phan et al., 2025). A game adds the requirement they leave out: the model has to recover its own state from raw perception, because nothing in the environment reports it. Game benchmarks for vision-language models each remove one part of that problem, whether perception, language, cultural prior or game knowledge (Paglieri et al., 2025; ARC Prize Foundation, 2026; Ju et al., 2026), so the combination stays untested.

We present JY-CRPG-BENCH, which places a vision-language model inside *Heroes of Jin Yong* (金庸群俠傳), a 1996 Chinese role-playing game run as its original DOS binary under emulation. Its ending requires recovering the fourteen books of the Jin Yong novels from an open world, a task that takes a first-time human player tens of hours. The model sees the 320×200 framebuffer and presses keys, through one fixed harness, so what varies between runs is the model. Three properties make the environment hard for current models, and Figure 1 shows them. The frame is 1996 pixel art carrying 16-pixel Traditional Chinese glyphs. The world is drawn in an isometric projection in which no key moves straight up the screen. The game keeps no quest log, draws no marker and reports no score, so beyond the opening route the brief supplies, deciding what is worth doing is part of the problem. ARC-AGI-3 places the same goal-setting requirement at the centre of agentic intelligence (ARC Prize Foundation, 2026). Progress is read from the memory of the emulated machine, from the save the game writes and from the published replay, none of which the model can call. It is summarised as a ladder of milestones that runs to the ending of the game.

We make three contributions.

1. We present JY-CRPG-BENCH, a long-horizon benchmark on a commercial open-world role-playing game whose difficulty comes from checkable properties of the environment, together with the brief, control surface and session protocol that every model plays under.

(a) the scene every session starts in
(b) the overworld the scenes hang off
(c) the four axes the game moves along
(d) dialogue, 16-pixel glyphs
(e) the menu inside a scene
(f) the record the benchmark reads
(g) the bag, one item described at a time
(h) the outdoor menu and the party
(i) a turn of battle on the grid

Figure 1: The environment at the resolution the model receives it, upscaled three times without interpolation: the two tiers of the world (a, b), the four axes the movement keys run along (c), the three kinds of text panel through which every objective, choice and statistic is stated (d–f), the bag (g), the outdoor menu with the party list it opens (h), and a turn of battle with the menu and record of the acting character (i).

2. We score progress against events of the game itself. Character records, the bag and the party are read from emulator memory and from the save the game writes, validated against the archives the release ships. The events the game keeps only on screen are read from the published replay against panels the game draws at fixed positions. Measurement never writes to the machine it observes, and every rung of the ladder is an event the model has to cause, up to the ending the game defines.
3. We evaluate 12 frontier models under the same agent harness and against a seeded random baseline under a wall-clock budget, and release the benchmark and every recorded session as a public, replayable leaderboard.

Table 1 places JY-CRPG-BENCH among interactive agent benchmarks. The next section reviews game environments as agent benchmarks and the measurement problem that long horizons create. Section 3 details the environment, the interface, the session protocol and the measurement, and Section 4 reports the field: how much of the ladder is open, where it separates the models, and how the budget is spent.

2 Related Work

2.1 Game Environments as Agent Benchmarks

Games have been the standard setting for sequential decision-making since the Arcade Learning Environment (Bellemare et al., 2013), and language and vision-language agents have produced a second generation of game benchmarks. BALROG aggregates six reinforcement-learning environments under one protocol and reports a persistent gap between what a model knows about a game and what it does inside one (Paglieri et al., 2025). lmgame-Bench isolates how much of a game re-

Table 1: Interactive agent benchmarks. *Observation* is what the model receives before any upscaling, *self-directed* marks environments that supply no task list, score or objective marker during play, and *progress from* is what a progress claim is checked against.

Benchmark	Environment	Observation	Language	Self-directed	Progress from
BALROG	6 research games	text, render	English	no	env. reward
TextQuests	interactive fiction	text	English	yes	game state
ARC-AGI-3	abstract puzzles	64×64 grid	none	yes	env. score
VideoGameBench	1990s action games	native px	English	no	checkpoint images
GameWorld	34 browser games	browser	English	no	game state
MineExplorer	Minecraft	3D render	English	no	milestone rules
JY-CRPG-BENCH (ours)	one 1996 CRPG	320×200 px	Trad. Chinese	yes	memory, save, replay

sult belongs to the harness (Hu et al., 2026), and Orak spans twelve popular titles with configurable agentic modules (Park et al., 2026). VideoGameBench is the nearest benchmark to ours, running real 1990s games from raw pixels under a no-scaffold rule (Zhang et al., 2025a). GameWorld standardises the action interface across 34 browser games and pairs every task with a state-verifiable metric (Ouyang et al., 2026). MineExplorer evaluates open-world exploration in Minecraft and finds that models handle single-hop tasks but degrade once hidden prerequisites must be coordinated over a longer trajectory (Ju et al., 2026); the opening of our environment has that structure. The nearest precedents in horizon are the public Pokémon runs. Claude 3.7 Sonnet reached three gym badges from screen pixels behind a memory scaffold, where an earlier model had not left the first house (Anthropic, 2025). Gemini 2.5 Pro finished Pokémon Blue in 813 hours behind a harness that translated the RAM of the game into text, and an ablation without vision performed about as well (Comanici et al., 2025). The PokeAgent Challenge turned the setting into a speedrunning track scored by completion time over milestones (Karten et al., 2026a). PokeGym builds a long-horizon benchmark on the open-world game Pokémon Legends: Z-A with no access to game state (Zhang et al., 2026a), and StarBench one on the turn-based Honkai: Star Rail (Zhang et al., 2025b). Crafter established the single-environment, achievement-based design we follow (Hafner, 2021), and NetHack remains the deepest environment in use for learned agents (Küttler et al., 2020). JY-CRPG-BENCH differs from these in running one persistent open world with Traditional Chinese as its only text, in verifying progress from emulator memory, the save file of the game and the published replay, and in scoring against the completion condition the game defines.

2.2 Long-Horizon Evaluation and Its Measurement

Vending-Bench established long-term coherence as the property that separates long-horizon tasks from long chains of short ones (Backlund & Petersson, 2025), and E-Commerce Bench extends the setting to a year of deterministic business operation (Fan et al., 2026). Both keep perception out of the loop. TextQuests shares our self-directed, long-horizon framing while removing vision (Phan et al., 2025), TextAtari stretches horizons to 10^5 steps over text renderings (Li et al., 2025). ARC-AGI-3 isolates adaptation to novel tasks by excluding language and external knowledge (ARC Prize Foundation, 2026), and AutumnBench probes world-model quality through environment-level queries (Warrier et al., 2025). UltraHorizon makes exploration under hidden rules the task itself (Luo et al., 2025), and Kapoor et al. (2026) observe that benchmark design has favoured tasks that are precisely specified, automatically graded and short, and call for open-world evaluation. Navigation from rendered images is a documented weakness: on mazes, one-shot navigation collapses for every model by ten cells a side (Jiang et al., 2026), and the failures are attributed to looping (Einarsson, 2025).

Long horizons strain measurement. The ABC checklist catalogues task-validity and outcome-validity failures across agentic benchmarks (Zhu et al., 2025). An audit of agent traces finds reward hacking in a majority of them on two suites (Shao et al., 2026), and a survey of the horizon gap finds that the field answers uninformative outcome signals by building denser step-level ones (Chen et al., 2026). State-based verification has precedent outside games, where τ-bench compares the final database state with an annotated goal (Yao et al., 2024). Our milestone ladder takes the same approach, read from machine state and from the recording so that dense signals cannot be mistaken for progress.

3 JY-CRPG-BENCH

3.1 ENVIRONMENT

Heroes of Jin Yong, published by Heluo Studio (河洛工作室) in 1996, opens with the protagonist in a small compound. The fiction supplies the only standing instruction: enter the world of the Jin Yong novels, join its factions, learn its martial arts, and recover the fourteen books that end the game. The world is two-tier, with many local scenes such as buildings, sects and caves attached to one outdoor world map of 480×480 tiles (scarsty & contributors, 2026). For each of 320 named characters the game maintains a record carrying level, experience, health, stamina, learned skills, carried items, reputation and aptitude. Progression is nonlinear and gated. Every scene carries an entry condition, and at the start all of them are closed except the home, five inns (客棧) and the house of the hermit (南賢); events open the rest, and the opening sets their order.

The compound holds a searchable chest, one speaking character and one exit tile. Beyond it, the path south leads to the hermit, whose long conversation opens the scenes the game keeps closed until then, the house of the first companion among them. A cabinet beside him holds the compass (羅盤), the only source of coordinates the game offers. The companion joins after a yes-or-no prompt at which the confirm key means no. Fights are turn-based on the scene grid. Each turn a character moves along the four axes and then strikes, uses an item, waits or rests. Skills draw on inner force (內力), and the first scripted fight opens by draining it. An automatic mode hands the character to the AI of the game, which alternates striking and resting and loses to an opponent who heals. A party that wanders elsewhere first meets storylines whose fights it cannot survive at level one; some of these fights end the game when lost, which then offers nothing but its own load menu, and others let the story continue.

Every objective, prompt and statistic reaches the model as Traditional Chinese rendered at sixteen pixels a line, and the character sheet on screen displays the same record the benchmark reads from memory. We selected this title for four reasons. Its text is Traditional Chinese throughout, a script that multimodal evaluation has largely neglected (Tam et al., 2025), so cross-lingual grounding is a central obstacle. It is menu-driven, so a model that deliberates between actions is not structurally disadvantaged as it would be in a real-time title (cf. Zhang et al., 2025a). Its progression state is rich enough to measure at the level of individual character statistics. Its $45°$ isometric projection makes the mapping from screen to action something the model must infer.

3.2 OBSERVATION AND ACTION SPACE

The model observes the raw framebuffer at 320×200 and acts by pressing keys. A scored session serves the native frame without upscaling, annotation, set-of-marks overlay or accessibility tree (cf. Xie et al., 2024). The frame sits at the hard end of several documented weaknesses of vision-language models. Downsampling costs them accuracy (Niu et al., 2025), degraded input costs them more (Usama et al., 2025), and images far from the training data degrade recognition even of familiar categories (Lin et al., 2026). Text rendered as pixels is understood less well than the same text supplied as tokens (Liu et al., 2026), and precise spatial reading fails once primitives are small or adjacent (Rahmanzadehgervi et al., 2024; Wang et al., 2024). The action space is the DOS keyboard, with 130 key names over 100 distinct keys, and Figure 2 shows how few of them the game answers. Movement runs along the four numpad diagonals, with the arrow keys aliased onto the same axes, so a model that presses `up` expecting to walk up the screen moves up and to the right. The view follows the character except near the edge of a scene, where the background holds still and the sprite walks across it, so a model that reads movement from the scrolling background loses count of its tiles there. Canopies and roofs are drawn over the character, and the entrance of a building is a single tile on one face of its wall, so reaching a door means walking around the compound to the side the door is on.

The one action presses a key, or a list of keys in order, holding each for a requested number of emulated frames, ten by default. It returns once the screen has reacted and held still, waiting through a scene transition if one starts, so the frame a model receives is the one to act on. There is no call that advances the game without input, since the game only moves on input, and the hold is counted in frames because the game samples its keyboard once per game-loop iteration. A scored session answers four of the calls in Table 2: the screen, the brief, the key names and the action. Three further

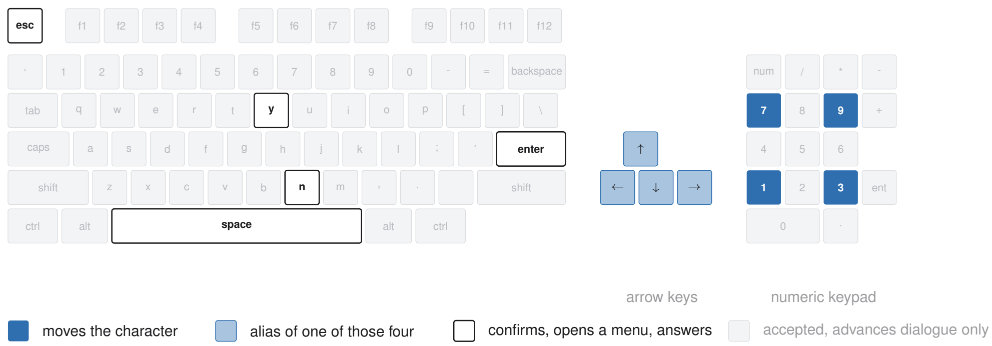


Figure 2: The keyboard the API exposes. The nine keys the game answers outside dialogue are picked out, the arrow keys are aliases of the four movement keys, and any key advances a dialogue.

Table 2: The control surface. A scored session answers the four calls marked; the three snapshot calls serve development.

Call	Body	Meaning	Scored
`GET /api/screen`	–	the current frame	✓
`GET /api/help`	–	the brief, in English or Chinese	✓
`GET /api/keys`	–	every accepted key name	✓
`GET /api/slots`	–	emulator snapshots on disk	
`POST /api/key`	`{"key":"kp3","hold":10}`	press one key, or several in order	✓
`POST /api/save`	`{"name":"..."}`	take a snapshot	
`POST /api/load`	`{"name":"..."}`	restore a snapshot	

calls save, list and restore emulator snapshots for development, and a scored session refuses them, so a run cannot be rewound from outside the game. The save and load menu of the game stays open to the model as to a human player, so a model that loses a fight can reload the last save the game wrote, and the recording shows it doing so.

3.3 Harness and Session Protocol

The benchmark supplies the brief and the control surface, and the model brings an agent. Every model in this paper ran inside pi[1], a minimal coding agent installed as shipped, whose four tools read, write and edit files and run a shell, with no web access, so no model could look up a walkthrough. The model drives the control surface with HTTP calls from that shell. The brief is the whole instruction the model receives: the calls that constitute play, the rules of a run, the opening route as far as the compass, and a manual of the controls and systems of the game. It is served in English and Chinese with the on-screen Traditional Chinese terms kept verbatim, so the language a model reasons in is not itself a handicap. An action returns only its metadata, so the model looks at the screen when it decides to, and the benchmark adds no memory scaffold, no map, no pathfinder and no access to the state of the game. Any memory a model keeps between actions it writes for itself in that shell. The leaderboard is therefore a ranking of models: harness choice moves game results (Hu et al., 2026; Karten et al., 2026b), and on long-horizon tasks the variance a harness induces can exceed the variance between models (Zhang et al., 2026b).

Table 2 lists the control surface in full, and the brief is published as `agents.md`, the project file pi reads at launch. A scored session is refused every field that would report its own progress until the run ends, so a model cannot query its own score.

A session is created under a declared name with a playtime, 60 minutes by default and up to twenty-four hours, and the clock starts at the first playable moment. The wall clock of every decision call is kept on the server, so played time and think time are measured for any client and shown beside a run without ranking it. Each session is a separate emulator process with a private copy of the game,

[1] `https://github.com/badlogic/pi-mono`

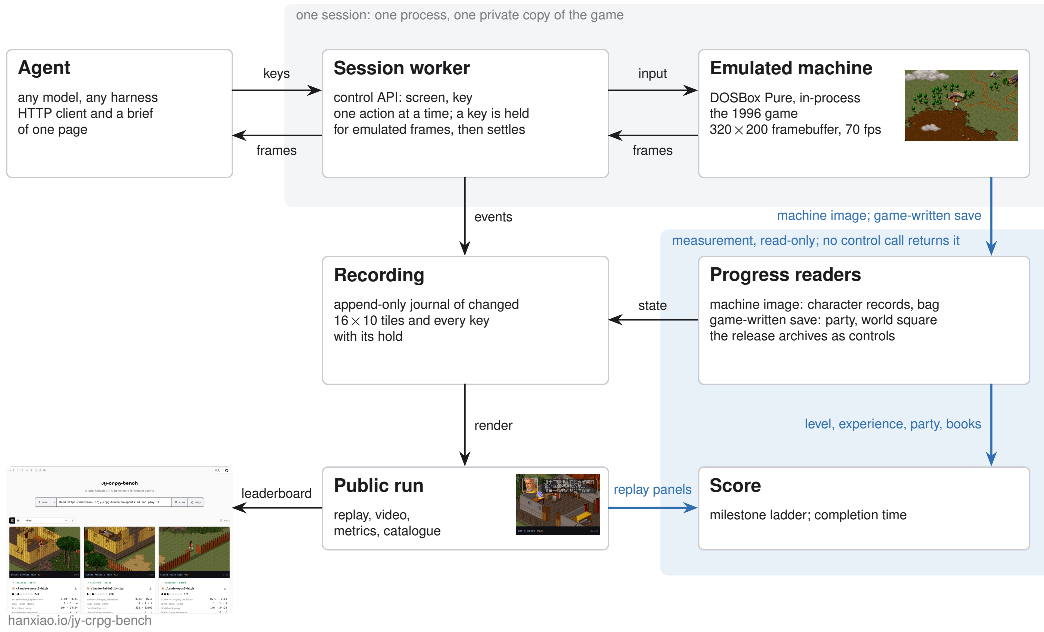


Figure 3: One process per session loads the game core and answers the control API; the recording feeds the public replay, and progress is read from the emulated machine, from the save the game writes and from the replay, on paths no control call reaches.

which every host supplies itself as described in Appendix B, with its save slots reset to a new game, so no run inherits the progress of another. A session ends only when its budget lapses or the game is completed, and there is no inactivity cutoff. When the budget lapses the session renders its recording to a time-compressed video, computes its metrics, and appends itself to the public catalogue with the video and its keypress timeline, so the behaviour behind a score can be reviewed. Figure 3 shows the apparatus, and Appendix A lists the emulation and hosting parameters.

3.4 Measuring Progress

The game keeps its persistent state in one archive: a base block with the world position, the party roster and the shared bag, followed by 320 character records of 182 bytes and an item table. We validated that layout against the opening machine image and the archives the release ships. We locate the same array in the live emulated machine by anchoring on one character name and confirming the two neighbouring records at the expected stride, since the name recurs in the image and the layout moves between runs. The bag in front of that array is the working copy and changes the moment an item is taken, so the item count, the compass and the books are read from it after every action and again from every save the game writes. A rung, once seen, is kept, so an item picked up and later spent still credits it. Reading is passive: the benchmark serialises the machine and never loads state into it. The only keys it presses on its own are those of the save described next, which take the game a few seconds of menu time that is not charged to the run and leave the screen as they found it.

For the party roster and the world position the benchmark reads the save the game writes. The game offers its save menu on the world map alone, and the menu itself shows where the party stands, since it has six rows on the world map and four in a scene. An attempt opens the menu, reads its height, and saves only where six rows appear. An attempt recurs every two minutes and once more before the budget lapses, and waits for a two-second gap in the keys the model sends. The session is held while the attempt runs, so the model never sees the screen mid-save, and the time of the first save that lands is the time of the crossing. Neither the archive nor anything read from it is reachable through the control surface, so a save that lands is itself the evidence that the party has crossed onto the world map. The session record also keeps a fingerprint of the world map taken from the frame, which credits the crossing only for a session with no save on record.

Five events leave no trace in either record, because the game keeps them only on screen. The model speaks with the hermit, opens the item screen (物品) with the compass in it, enters a fight, loses it, or answers the prompt of a companion who asks to join. These are read from the published replay. Each event draws a panel at a fixed position of the screen. The hermit has a portrait in the dialogue frame, and the compass adds a coordinate line to the item screen. A fight shows the card of the acting character, a defeat shows its banner (戰鬥失敗), and a recruitable character asks with a yes-or-no prompt. The panels are fixed art at fixed places, and every frame of the replay is matched against them at one frame a second, with a match above 0.9 counted as a hit. A prompt followed by a yes key in the keypress journal is a recruitment.

Progress is summarised as 10 milestones, in the spirit of the achievements of Crafter (Hafner, 2021). The rungs fall into two phases that the game itself separates. The opening needs no fight. Its five rungs are the chest in the first room, the exit tile, the conversation with the hermit, the compass in his cabinet, and the companion who joins through dialogue, all reached by walking, reading and answering. The campaign begins with the first fight, which a party at level one can lose, and runs through the experience and levels that victories pay and the chains of items, factions and fights that the books sit behind. Its five rungs are a fight entered, a fight fought to its end, experience gained, the second level, and the first of the fourteen books. The default budget is sized to the opening and the extended budgets to the campaign. The rungs are listed in the order the opening imposes, but a run is credited with every rung it reaches, so the count does not depend on the path taken.

The ladder runs to the ending of the game. The fourteen books, named in Appendix C, are what the game requires before its final sequence can be played, and the archive records every book held. The count of books therefore extends the ladder beyond its last rung, and the first read that finds all fourteen held records the completion time. A session that completes carries that time, which orders a field in which every entrant passes, so one instrument covers both phases and the benchmark keeps measuring after the first model finishes.

The ladder is the fixed score of the benchmark. For the multi-hour budgets the protocol allows, the release adds a second, additive report over the same ground truth. It records a finer trajectory, the scenes a run reached, the battles it survived, the skills it learned and the books it gathered, out to the ending, with the hard prerequisites of the walkthrough as dependency gates between them. Each component is scored against a fixed denominator with its coverage reported beside it, and the report never rescores the ladder, so a long run holding no book still leaves an ordered record of how far its plan carried. Its checkpoint contract and component weights are released with the benchmark.

3.5 Random Baseline

A seeded random policy plays the same budget through the same control surface, drawing from 13 keys the brief teaches, weighted towards movement and confirmation, and never reading the screen. This follows the convention that anchors a score on random and human play (Mnih et al., 2015; Hafner, 2021; ARC Prize Foundation, 2026). The rungs it reaches mark the progress available without reading the screen and are the floor of the ladder. Because it is seeded, its whole key sequence is recomputed offline and checked against the journal of keys the machine received.

4 Evaluation

The field at submission is 12 models from 5 vendors at the 60-minute budget, and a seeded random baseline. Every session a model in the field played at this budget is on record, 42 in all and 2 to 6 per model. A model is credited with every rung any of its sessions reached, with the number of sessions shown beside it, and a session created under a variant spelling of a model name is listed under the model. Every session ran in pi under the same settings, and effort is reported as wall clock. Every session is public with its video and replay timeline, and the session records the paper is generated from are committed beside the paper source. The subsections that follow report how much of the ladder is open, where it separates the models and what the replays show at each division, and how the budget is spent.

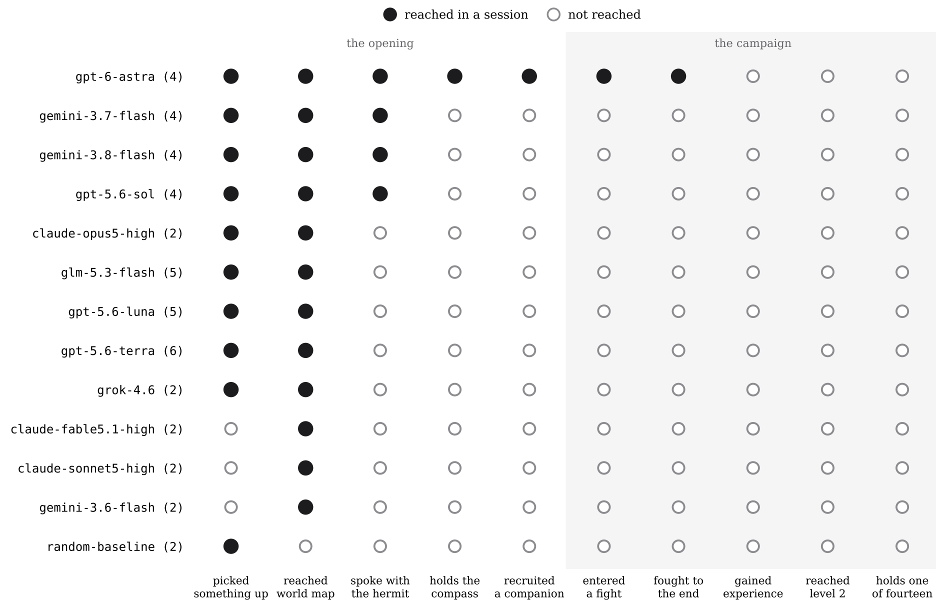


Figure 4: The milestone ladder, one row per model with the number of sessions it played in brackets; a filled marker is a rung reached in at least one of those sessions. The evidence behind each rung, and the record it is read from: an item count above its opening count (the bag in memory); the save the game writes on the world map, or the world-map fingerprint of the frame (save, frame); the portrait of the hermit in the dialogue frame (replay); the compass in the bag or its coordinates on the item screen (bag, replay); a party of more than one or the prompt of a companion answered yes (save, replay); the card of the acting character in a fight (replay); the defeat banner or experience gained (replay, record); experience above zero and a level above one (character record); a book in the bag or carried by the protagonist (bag, record).

4.1 Headroom

Of the 12 models, 9 pick something up, all 12 reach the world map, 4 speak with the hermit, and one, `gpt-6-astra`, takes the compass, recruits a companion, enters a fight and fights it to the end. It reaches 7 of the 10 rungs across its 4 sessions. No model gains experience: every one of the 37 character records on record remains at level 1 with 0 experience and 1 skill, as it was when the game began, so every rung from experience on is open. Figure 4 shows the ladder by model. Comparable benchmarks report their fields at a similar point: BALROG places its best model at 1.5% progression on NetHack (Paglieri et al., 2025), and VideoGameBench reports the best vision-language models completing 0.48% of its suite (Zhang et al., 2025a). The distance from the first fight to the fourteen books is the headroom the benchmark holds.

4.2 Discrimination

The chest, the hermit and the fight divide the field, each differently, and Figure 5 collects one frame from each pattern behind the divisions. The random policy reaches 1 of the 10 rungs in 60 minutes: in a small room a random walk searches the chest within the hour. Its recomputed key sequence matches the journal of the machine key for key, and no save lands during its hour, so the world map is the first rung a random walk does not reach. The chest therefore separates models from one another but not from the floor: 9 models searched it and 3 never did, so the count of rungs summarises what a model achieved and presumes no single path.

Every model crosses onto the world map, and the replays show what the crossing costs. The compound is left through a yard closed by a fence with one gate, and the yard held several sessions. `glm-5.3-flash` needed 1595 actions to cross, against a median of 86, and reached the world map at minute 37 after long stretches pressed into the fence. It never pressed enter or space in that session, so the chest in the first room, which opens on either, stayed shut. Beyond the fence the hour goes to the map itself. `claude-fable5.1-high` crossed at minute 13 and then spent 1261 actions and 44 minutes on the world map without entering a scene, and passed the gate of the hermit repeatedly. The entrance of a walled compound is one tile on one face, and every other face is solid. Two sessions that found an inn and a shop talked to the keeper repeatedly, and nothing in the reply advances the opening. `gpt-5.6-terra` sent lists of up to 100 keys in one action, 22 on average, with 32 actions of fifty keys or more, so it looked at the screen once for every 22 keys it pressed. It reached the world map, credited by the save alone. The world-map rung also shows why the save is read where the game writes one. Of the 10 crossings that carry both a save and a screen fingerprint, the two agree on 9, the save credits 1 crossing the fingerprint missed, and the fingerprint never appears without a save behind it.

The hermit divides the field a second time, and he divides it on reading. This rung and the fights and recruitment above it are read from the replay, and no frame without the panel scored above 0.79 against the threshold of 0.9. `gemini-3.7-flash`, `gemini-3.8-flash`, `gpt-5.6-sol` and `gpt-6-astra` reach his conversation, in 8 sessions, and the conversation is long. `gemini-3.8-flash` pressed through his 40 lines at 23 seconds each, so it took 15 minutes and ended at minute 54, when the hermit had named the cabinet and 6 minutes of budget remained. `gemini-3.7-flash` reached him at minute 37 and pressed 60 times in 1 minute. Neither searched the cabinet. The 4 sessions of `gpt-6-astra` all reached him between minutes 11 and 19. The session with the fullest record pressed through the conversation in 7 minutes at 10 seconds a line, searched the cabinet, and then opened the item screen 9 times to read its coordinates from the compass, the one instrument the brief recommends. Its 42 arrow-key presses all fell inside menus and prompts, never on the map, and it entered 11 scenes in the hour.

Above the compass the field is a single model. All 4 sessions of `gpt-6-astra` read the compass between minutes 18 and 29, and from there they diverged. One met Duan Yu (段譽) at an inn at minute 46, answered yes at minute 52 when he asked to travel together, and was handed the manual of his art (六脈神劍譜). One entered a fight at minute 27 and lost it at minute 28, then opened the system menu (系統), loaded the save the benchmark had written, and by minute 34 held the compass again. One entered a fight at minute 40 and sent its last key at minute 42 with the fight still open. The fight divides the field a third time, and there the field ends: the rungs of the campaign beyond it are undivided.

4.3 Budget and Effort

Of the 42 sessions, 31 played to their budget, and 11 ended early. The crossing onto the world map is timed by the game, since the first save it writes there trails the crossing by at most the two-minute cadence of the attempts. By that clock the 10 crossings whose save carries a time fall between 9 and 41 minutes into the session, with a median of 26. One session of `gpt-6-astra` crossed at 9 minutes by that clock, after 33 actions, and spent the rest of the hour beyond the exit. The hour therefore covers the first room and, for one model, the walk south to the cabinet and beyond. The budget is wall clock, so the compute a model spends on a decision is charged against the time it has left to play, an efficiency framing shared with ARC-AGI-3 and TextQuests (ARC Prize Foundation, 2026; Phan et al., 2025). The sessions of the field took between 31 and 1994 actions, and the same conversation with the hermit cost one model 10 seconds a line and another 23. A comparison of token spend across models waits for a field in which more than one model plays past the opening. A model that has not reached the hermit has not begun the game the ladder scores, and at this budget `gpt-6-astra` alone has, so a cost per rung would rank one model.

5 Conclusion

We presented JY-CRPG-BENCH, a long-horizon benchmark that runs a 1996 open-world role-playing game, gives every model the same brief and control surface, and scores it against events of the game itself. Progress is read from the memory of the machine, from the save the game writes and from the published replay, and the model can call none of it. At the 60-minute budget the field of 12 models stays inside the opening of the game: every model reaches the world map, 4 speak with the hermit, and one, `gpt-6-astra`, takes the compass, recruits a companion and fights. Every rung from experience to the fourteenth book stands open, and the one rung the random baseline reaches marks the floor a model has to clear. That a single model plays past the opening while the rest of

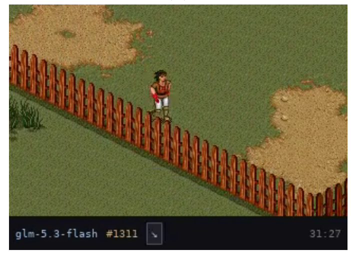
(a)

(b)

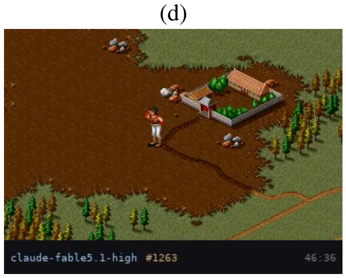
(c)

(d)

(e)



(f)



(g)

(h)

(i)

Figure 5: Frames from the published replays, each with the strip of its video naming the model, the action count and the clock: (a) `gpt-6-astra` reads its coordinates from the compass on the item screen and (b) presses through the conversation of the hermit; (c) the same model at the prompt of Duan Yu in an inn, (d) in a fight, and (i) at the defeat banner; (e) `gemini-3.8-flash` is told where the compass lies; (f) `glm-5.3-flash` in the yard behind the fence; (g) `claude-fable5.1-high` passing the gate of the hermit; (h) `grok-4.6` talking to an innkeeper.

the field does not measures the environment. It is hard in ways that are named and checkable: the frame is a 320×200 isometric render of 1996 pixel art, objectives are stated only in Traditional Chinese, and the world supplies no score. The ladder runs to the ending of the game: the archive records the fourteen books held, so a model that finishes leaves proof of it. From that point the leaderboard ranks by completion time, with the video and every key of the finishing run on record. Three fields follow this one. Sessions of four to twenty-four hours bring the campaign rungs and a completion time within reach and make a token cost per rung a comparison between models. The same models with web access measure what a walkthrough is worth against the brief alone. Once a model finishes, the completion clock defines the speedrun.

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A Session and Infrastructure Parameters

A host of 8 vCPUs and 16 GiB runs up to 24 concurrent sessions. Emulation is DOSBox Pure through libretro at a fixed 26800 cycles, the speed of a 486DX2-66 and period-correct for the game, with no sound card, so the observation is visual only. The published video runs at 8 times speed or faster for long runs, with a strip above the frame naming the actor, the action number, the keys held and the clock, and a timeline file beside it carries every keypress and its hold duration for the replay.

B Release and Licensing

The benchmark code is released under MIT. The emulation core (`dosbox_pure_libretro`) is GPLv2, loaded at runtime through the libretro C API and redistributed unmodified. The bundled macOS core is built from `schellingb/dosbox-pure` at `7f6e8fb`, and the hosted service ships a Linux build pinned by image digest. The game is the 1996 commercial release, copyright Soft-World International (智冠科技) and Heluo Studio. It is not redistributable, it is not tracked in the repository, and every host supplies its own copy. The public catalogue therefore carries only the metrics, videos and replay timelines we measured, and nothing that substitutes for the game data. The frames of the game in this paper identify the interface under study, in the same spirit as prior work showing frames of copyrighted games (Paglieri et al., 2025); if the rights holders object, they will be removed.

C The Fourteen Books

The ending requires the 14 books, which the game stores as item records 144 to 157 of its item table, named after the novels: 飛狐外傳、雪山飛狐、連城訣、天龍八部、射鵰英雄傳、白馬嘯西

風、鹿鼎記、笑傲江湖、書劍恩仇錄、神鵰俠侶、俠客行、倚天屠龍記、碧血劍、鴛鴦刀. A book counts as held when it is in the shared bag or in the four carried slots of the protagonist. With all fourteen in hand and sufficient reputation, the game opens its final sequence at a shrine on the far side of the map, where the books are placed and a last battle decides the ending (Bahamut forum contributors, 2023).